KATHFORD INTERNATIONAL COLLEGE OF ENGINEERING AND MANAGEMENT
DEPARTMENT OF COMPUTER AND ELECTRONICS COMMUNICATION & INFORMATION ENGINEERING

A
MINOR PROJECT REPORT
ON
“NIGHT PATROLLING ROVER”

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Lalitpur, Nepal

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KATHFORD INTERNATIONAL COLLEGE OF ENGINEERING AND MANAGEMENT
BALKUMARI, LALITPUR

(NIGHT PATROLLING ROVER)

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A PROJECT WAS SUBMITTED TO THE DEPARTMENT OF COMPUTER AND ELECTRONICS, COMMUNICATION & INFORMATION ENGINEERING IN PARTIAL FULLFILLMENT OF THE REQUIREMENT FOR THE BACHELOR'S DEGREE IN ELECTRONICS, COMMUNICATION & INFORMATION / COMPUTER ENGINEERING

DEPARTMENT OF COMPUTER AND ELECTRONICS, COMMUNICATION & INFORMATION ENGINEERING
LALITPUR, NEPAL

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KATHFORD INTERNATIONAL COLLEGE OF ENGINEERING AND MANAGEMENT

BALKUMARI, LALITPUR

DEPARTMENT OF COMPUTER AND ELECTRONICS, COMMUNICATION & INFORMATION ENGINEERING

The undersigned certify that they have read, and recommended to this department for acceptance, a project report entitled "Night Patrolling Rover" submitted by Ankit Mandal, Bibek Khanal and Bhuwan Adhikari in partial fulfilment of the requirements for the Bachelor's degree in Electronics & Communication / Computer Engineering.

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Supervisor, name of Supervisor

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Name of the Organization, he/she belongs to

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Coordinator, Name of Coordinator

Title

Name of the coordinating committee

DATE OF APPROVAL:

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Departmental Acceptance Letter

DEPARTMENTAL ACCEPTANCE

The project entitled "**Night Patrolling Rover**", submitted by Ankit Mandal, Bibek Khanal and Bhuwan Adhikari in partial fulfilment of the requirements for the Bachelor's degree in Electronics, Communication& Information / Computer Engineering has been accepted as a bonafide record of work carried out by them in this department.

Name of department head

Title

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Thank you all for being a part of this endeavor.

Ankit Mandal
Bibek Khanal
Bhuwan Adhikari

ABSTRACT

The "Night Patrolling Rover" project focuses on the development of an autonomous Rover equipped with advanced sensors for enhanced patrolling capabilities during night-time operations. The Rover integrates PIR sensor for motion detection, and ultrasonic sensor for obstacle detection to navigate safely and efficiently in various environments. Key components of the Rover include an array of sensors pir sensor for motion detection, and ultrasonic sensors for real-time obstacle detection and avoidance. These sensors collectively enable the Rover to autonomously patrol designated areas, ensuring security patrolling without human intervention.

The primary objective of this project is to demonstrate the feasibility and effectiveness of utilizing multiple sensor technologies within a single autonomous platform. The rover's autonomous navigation system leverages data from ultrasonic sensors to detect obstacles, thereby enhancing its operational safety and effectiveness in low-visibility conditions typical of night-time patrols.

The Rover's autonomous navigation system utilizes data from the ultrasonic sensors to detect obstacles in its path thereby enhancing its ability to operate safely and effectively in low-visibility conditions typical of night-time patrols. This project aims to demonstrate the feasibility and effectiveness of integrating multiple sensor technologies into a single autonomous platform for night patrolling, contributing to advancements in autonomous Rover systems for patrolling and security applications.

Keywords: Patrolling, Rover, Detection

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1. INTRODUCTION

1.1 Background

In recent years, there has been a pressing need for enhanced security measures and environmental monitoring, particularly during nighttime hours when visibility is limited, and potential risks are heightened. Traditional methods of patrolling and surveillance often rely heavily on human presence, which can be costly, inefficient, and subject to human error.

In response to the increasing demand for effective nighttime patrolling and monitoring, the ‘Night Patrolling Rover’ project introduces an autonomous Rover equipped with advanced sensor technologies. The Rover integrates key sensors including motion detectors for security purposes, and ultrasonic sensors primarily for obstacle detection and avoidance.

The development of autonomous rovers is not only a response to the limitations of traditional patrolling methods but also a reflection of advancements in robotics and sensor technology. Recent innovations in artificial intelligence, machine learning, and Internet of Things (IoT) have paved the way for creating intelligent systems capable of making real-time decisions and adapting to dynamic environments.

Rover’s autonomous capabilities leverage ultrasonic sensors to navigate safely through various terrains and conditions, detecting obstacles in real-time. This technology enables the Rover to autonomously patrol designated areas, enhancing patrolling effectiveness and reducing the reliance on human intervention during night-time operations.

This project addresses critical needs in security and monitoring, aiming to deploy a robust and autonomous platform capable of conducting efficient patrols in low-light environments.

1.2 Problem Statement

The Night Patrolling Rover aims to tackle the challenge of enhancing nighttime patrolling using autonomous Rover technology. Traditional methods of night-time patrols often rely on human presence, which can be inefficient, costly, and prone to human error. This project seeks to develop a solution that utilizes advanced sensors such as PIR sensor, and ultrasonic sensors to enable a Rover to autonomously navigate and patrol designated areas.

Older rover systems face additional challenges due to outdated technology and design limitations:

1. **Limited Functionality**: Older rover systems may lack advanced features such as real-time data processing, adaptive navigation algorithms, or remote monitoring capabilities. This limits their effectiveness in meeting current security monitoring demands.

2. **High Maintenance Requirements**: As components age, older rover systems may require increasingly frequent maintenance and repair. Finding replacement parts or upgrading outdated components can be challenging and costly.

   Key issues to be tackled include:

   1. **Autonomous Navigation:** Developing rover to autonomously navigate in low-light conditions using ultrasonic sensors for obstacle detection.
   2. **Sensor Integration:** Integrating multiple sensors for comprehensive environmental and security monitoring.
   3. **Reliability and Efficiency:** Ensuring the rover operates reliably with minimal downtime and maintenance requirements.
   4. **Security and Patrolling Effectiveness:** Enhancing the rover’s capability to detect and respond to potential threats during patrols.

1.3 Objectives

The aim of this project is to develop an efficient, reliable, and effective autonomous night patrolling system that can significantly enable better responses to critical missions and contribute to safer and more resilient operations.

1.3.1 Main Objective

i. The main objective of this project is to develop a versatile Rover for patrolling in various locations with alert assistance features.

1.3.2 Specific Objective

i. To develop an autonomous Rover capable of moving around.
ii. To detect suspicious motion of humans at the patrolling location.
iii. To capture the video in real time and send it to monitoring unit.

1.4 Scopes

Developing an autonomous Night Patrolling Rover with advanced sensors for enhanced patrolling capabilities during nighttime operations.

- **Border Security:** The autonomous rover could be used for patrolling and monitoring border areas to enhance security and detect any unauthorized activities.

- **Industrial Surveillance:** The rover can be deployed in industrial settings to monitor equipment, detect anomalies, and ensure safety and security in the workplace.

- **Smart Cities:** Integrating the rover into smart city infrastructure could improve urban surveillance and public safety by patrolling streets, monitoring and detecting potential security threats.

2. LITERATURE REVIEW

Before starting this project, we spent a lot of time exploring what's already been done in the field of security robots, especially focusing on night patrolling robots. Our goal was to understand what technologies work best, what common challenges exist, and how we could design a system that's both effective and practical for real-world use. The following review covers the most relevant research papers, all published within the last few years, which helped shape our decisions.

One of the first papers we looked at was by **Pindoo et al.** [1], who built a night vision patrolling robot using a Raspberry Pi. Their robot followed a fixed path and used both a night vision camera and sound sensors to detect disturbances. If it detected a sound, the robot would turn toward it, capture an image, and send that image to the user via Blynk IoT platform. What we liked about this work was its simplicity and use of affordable hardware, which made it feel very achievable for us.

Another interesting study, from **Bhatia et al.** [2], focused on women's safety using a night patrolling robot. Their design combined ultrasonic sensors for obstacle detection, PIRsensors to sense movement, and a night vision camera to capture images. What stood out here was their use of GSM to send instant alerts to a control room, along with live video streaming via NodeMCU. This combination of real-time communication and video streaming directly inspired how we approached the communication part of our project.

Of course, while following a line, the robot still needs to handle unexpected obstacles. That's where **Chacko et al.** [5] provided useful insights. Their robot could detect obstacles using ultrasonicsensors, pause, capture an image, and then send that image to a remote user. This blend of obstacle detection and evidencecapture fit well with what we wanted to achieve.

Since night patrolling relies heavily on visual monitoring, we also explored computer vision techniques. Both **Bhatia et al.** [2] and **Chacko et al.** [5] used OpenCV for basic image processing. If we wanted more advanced object detection, research by **Redmon and Farhadi** [7] on the YOLO(You Only Look Once) model showed how even embedded platforms like Raspberry Pi could run real-time object detection with reasonable performance. This convinced us to consider OpenCV for basic processing and YOLO for future upgrades.

Finally, we wanted to make sure our robot could immediately alert the user if something suspicious happened. Both **Bhatia et al.** [2] and **Pindoo et al.** [1] used SMTP (Simple Mail Transfer Protocol) for sending images via email, and Blynk for sending notifications. This approach seemed practical and lightweight, making it a good fit for our project as well.

After reviewing all these papers, it became clear that the best approach for our night patrolling robot would be a hybrid system combining ultrasonic sensors for obstacle detection, PIR sensor for human or animal intervention, and a night vision camera for 1visual monitoring using Rpi-Cam-Web-Interface. Real-time alerts would be handled by Server (PC GUI) via a TCP socket. By blending ideas from these existing works, we could build a cost-effective yet reliable patrolling robot that’s suitable for small premises or indoor use.

3. PROJECT METHODOLOGY

The project methodology for developing the Night Patrolling Rover involves four key steps:

1. **Concept Design:** This initial phase involves conducting various studies to identify shortcomings in traditional patrolling systems and explore advanced methods in environmental monitoring.

2. **Equipment Selection and Creation:** In the second step, the necessary physical equipment is chosen. This may involve selecting sensors, hardware components, and other necessary tools to equip the rover.

3. **Software Development:** The third phase focuses on creating the software system for the rover. This includes programming instructions for the hardware components and designing a web application to track and analyze environmental data collected by the rover.

4. **Testing:** In the final testing stage, the Rover is deployed to gather detailed information about the area it operates in. This phase allows for the evaluation of the rover’s performance, functionality, and effectiveness in autonomously patrolling and monitoring the environment.

3.1 Block diagram of Developed System

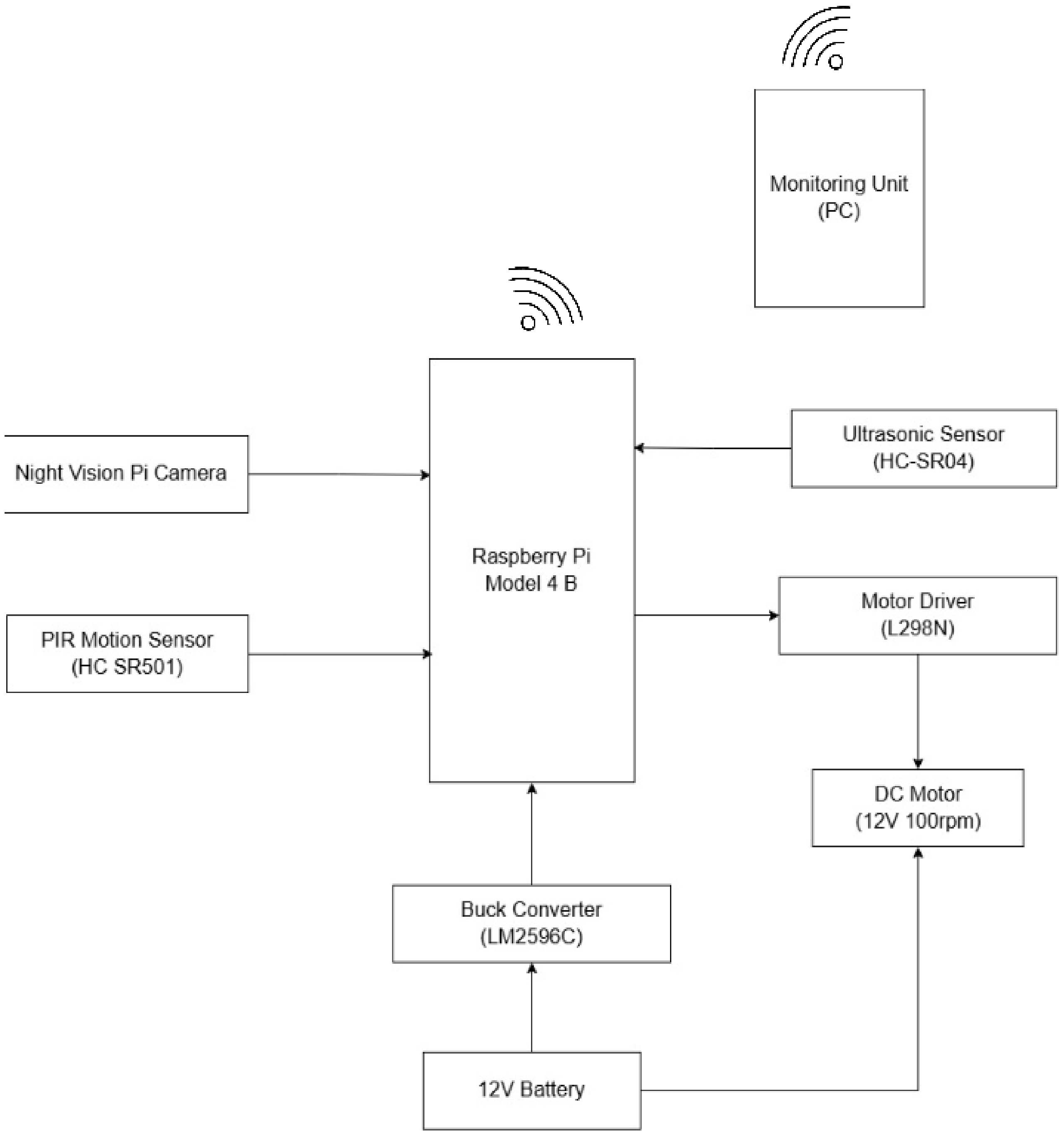

Figure 4.1: Block Diagram of Rover System

Brief overview of the components and their connections:

i. Raspberry Pi (PI): The central unit that controls the system.
ii. Night Vision Pi Camera: Connected to the Raspberry Pi for capturing images or video, especially in low light conditions.
iii. PIR Sensor (HC SR501): Connected to the Raspberry Pi to detect motion.
iv. Ultrasonic Sensor (HC SR04): Connected to the Raspberry Pi for measuring distance.
v. Motor Driver (L298N): Controlled by the Raspberry Pi to drive the DC Motor.
vi. DC Motor (12V 100RPM): Connected to the Motor Driver for actuation purposes.
vii. Monitoring Unit (PC): Receives data from the Raspberry Pi via Wi-Fi for remote monitoring.
viii. 12V Battery: Provides power to the system.
ix. Buck Converter (LM2596C): Converts the 12V from the battery to a lower voltage suitable for the Raspberry Pi and other components.

3.2 Flowchart of Developed system

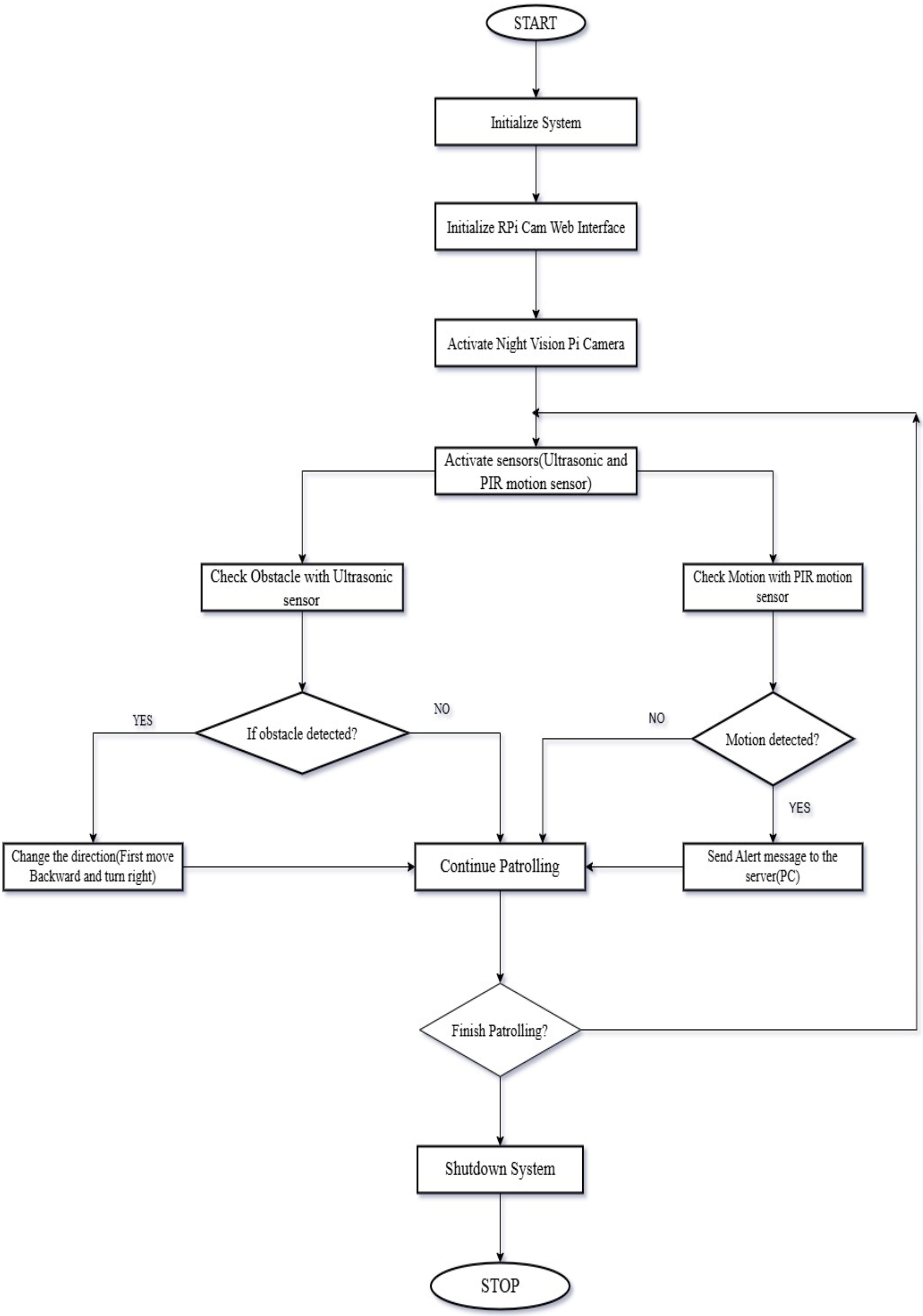

Figure 4.3: flowchart of developed system

4. HARDWARE AND SOFTWARE REQUIREMENTS

4.1 Hardware:

i. Rover Chassis

The base structure that houses all components, providing stability and mobility for the night patrolling rover.

ii. DC Motors and Wheels

Enables movement and navigation of the rover. The motors drive the wheels, allowing controlled motion.

iii. Raspberry pi

A single-board computer that serves as the brain of the rover, handling sensor data processing and communication.

iv. PIR Motion sensor (SP503)

Detects human or animal movement using infrared radiation, allowing the rover to identify intrusions.

v. Buck Converter (LM2596C)

A voltage regulator that steps down the power supply to a safe operating level for different electronic components.

vi. 12V LiPo Battery

The main power source for the rover, providing energy to all electronic components.

vii. Night Vision Pi Camera module

Captures real-time video and images in low-light or nighttime conditions for surveillance.

viii. Motor Driver(L298N)

Controls the speed and direction of the DC motors by regulating power supply.

ix. Ultrasonic sensor (HC-SR04)

Measures distance to obstacles using sound waves, helping in obstacle avoidance.

4.2 Software:

i. Proteus

A Simulation Software used for designing and testing electronic circuits before actual implementation.

ii. Raspbian

A Linux-based operating system optimized for Raspberry Pi, used to run the rover’s software.

iii. Blender

A 3D modeling tool used for designing and simulating rover components.

iv. Python

The primary programming language used for controlling sensors, motors, and data processing.

v. RPi-Cam-Web-Interface

A web-based interface for streaming live video feed from the rover’s camera, enabling remote monitoring.

5. RESULTS AND DISCUSSION

The developed autonomous night patrolling rover successfully met the objectives outlined for the project, demonstrating efficiency in navigation, motion detection, and real-time surveillance. The rover was tested in various environments to assess its mobility, obstacle avoidance capabilities, and response to different terrains. The results showed that the system effectively navigated predefined paths and dynamically adjusted its movement to avoid obstacles using integrated sensors such as the ultrasonic and PIR sensors.

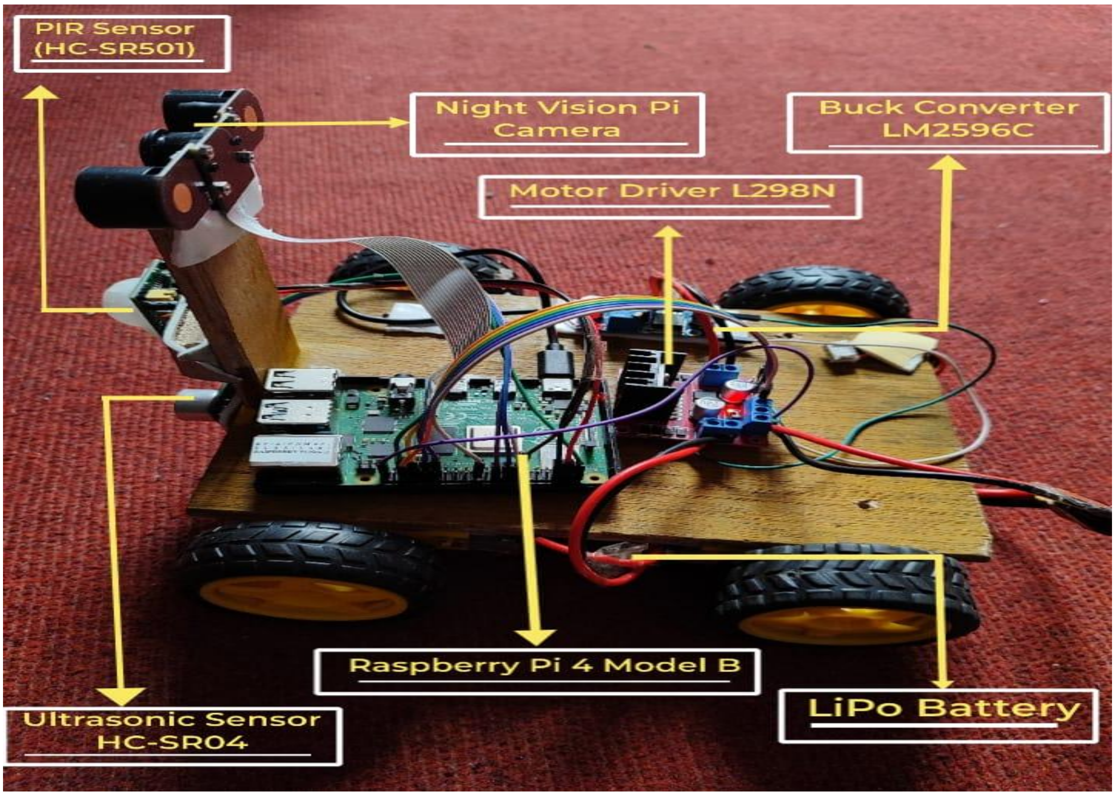

Figure 5.1: Developed Model of Rover

The motion detection system performed reliably, accurately identifying human and animal movements in the patrolling area. The PIR sensor, combined with video processing techniques, enabled the system to detect motion and trigger alerts in real-time. This ensured immediate attention to any unusual activity detected during the patrol. Additionally, the system successfully captured and transmitted real-time video footage to the monitoring unit, providing continuous surveillance.

The seamless video transmission ensured that security personnel could monitor the area efficiently and take appropriate action when necessary.

Moreover, the implemented alert mechanism effectively notified the monitoring unit upon detecting suspicious movement, enhancing the system's security response. The combination of motion detection, real-time video streaming, and alert notifications made the rover a reliable tool for night patrolling, reducing the need for manual work. The overall performance of the system demonstrated its potential for real-world applications in areas requiring continuous surveillance and security enforcement.

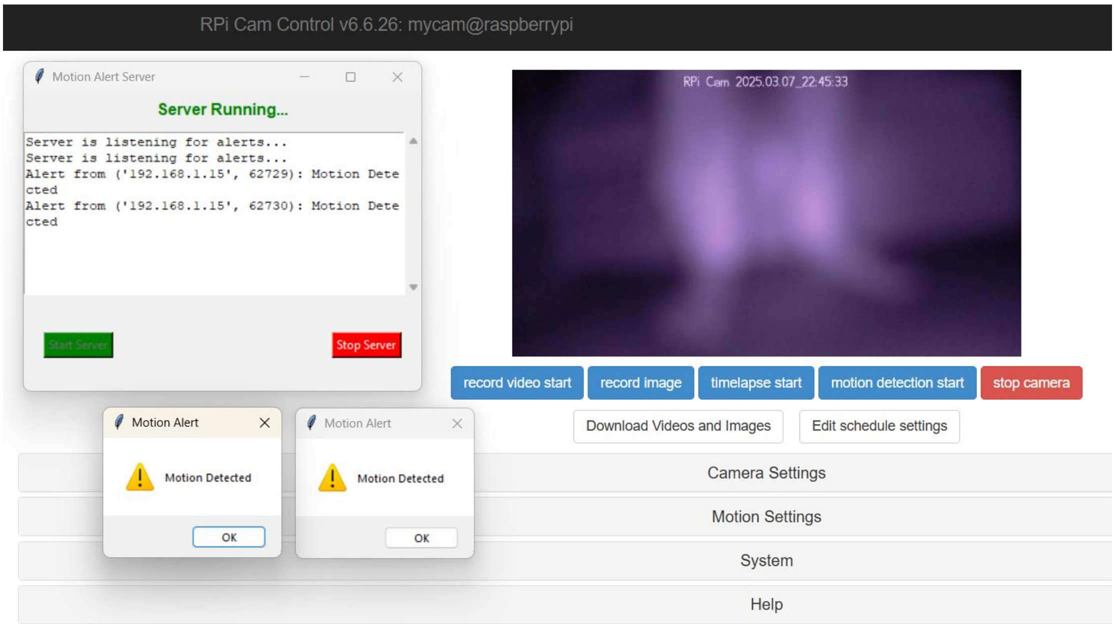

Figure 5.2: Graphical Interface for Live Monitoring and Alert System

The test of the rover was conducted in the dark room within the premise of kathford college and the results of this test confirm that the developed system meets the intended objectives and provides a solution for autonomous night patrolling. The rover’s ability to navigate autonomously and detect motion effectively aligns with the findings of previous studies on robotic surveillance systems. Traditional security systems, such as static CCTV cameras and manually operated patrols, often face limitations in mobility and real-time response. In contrast, this project offers an improved approach by enabling autonomous movement, reducing the need for human involvement, and enhancing overall surveillance efficiency.

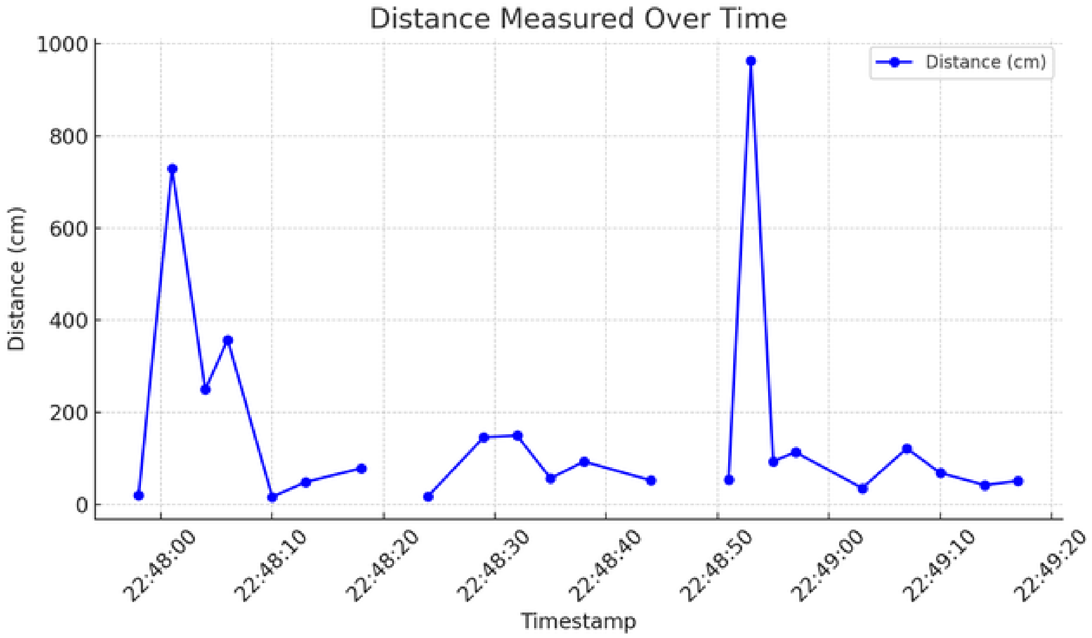

Figure 5.3: Obstacles Distance Over Time

The first graph illustrates how the rover detected obstacles at different distances. Some obstacles were very close, requiring quick avoidance, while others were farther away, allowing smoother navigation. The data confirms that the system effectively detects and responds to objects in its path.

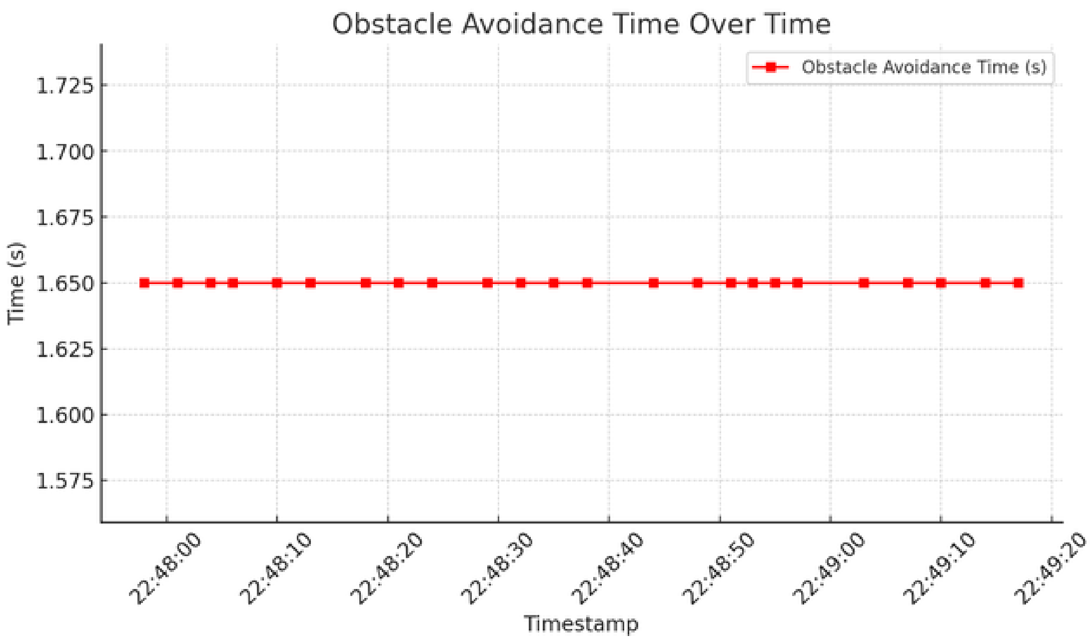

Figure 5.4: Obstacles Avoidance Time Over Time

The second graph highlights the time taken for the rover to avoid obstacles. The relatively steady values suggest that the rover consistently maneuvered around objects without major delays, showing the reliability of its navigation system.

Comparing the system's performance with existing literature on autonomous surveillance systems, this project demonstrated advancements in real-time motion detection and alert generation. Studies on security robotics have highlighted the significance of motion detection using PIR sensor. The results obtained in this project align with these findings, confirming that a combination of PIR sensors and video processing enhances motion detection accuracy. Additionally, research on surveillance technology often points to challenges in real-time data transmission. While some existing systems struggle with delayed video streaming, this project achieved low-latency video transmission, allowing instant monitoring and improving security response time.

6. CONCLUSION

This project successfully developed an autonomous night patrolling rover designed to enhance security through real-time surveillance, motion detection, and automated alerts. The rover was able to navigate independently, detect human and animal movements, and transmit live video footage to the monitoring unit. Through multiple tests, it demonstrated reliable movement, accurate motion detection, and smooth real-time video streaming with minimal delay, making it an effective solution for night patrolling.

One of the most important achievements of this project is the rover's ability to patrol autonomously, reducing the need for manual security personnel and improving response times to potential threats. With the integration of sensors like PIR sensor and ultrasonic sensor, the system effectively identified movement while avoiding obstacles. The real-time alert mechanism added another layer of security, ensuring that any unusual activity was immediately reported.

In conclusion, this project is a step forward in autonomous security technology, offering a practical and innovative approach to night patrolling. With further refinement, it has the potential to be used in industrial zones, military bases, and public safety applications. The insights gained from this work contribute to the growing field of autonomous surveillance.

7. LIMITATIONS AND FUTURE WORK

While the autonomous night patrolling rover proved to be effective, there are a few limitations that could be improved upon in future versions. One challenge was the sensitivity of the sensors to environmental factors such as lighting, temperature, and changes in terrain. In certain conditions, the motion detection was less accurate, and the rover struggled to differentiate between normal movement and potential threats, leading to some false alarms.

Another limitation was the rover's ability to navigate in more complex environments, especially those with uneven terrain or narrow spaces. While the system performed well in most situations, it could benefit from enhanced capabilities to handle these more challenging environments. The quality of video streaming was also another field that can be improved by using high quality camera with greater number of megapixel.

For future work, integrating AI-powered image recognition could significantly improve the rover's ability to distinguish between routine activities and suspicious behavior, reducing the chances of false alarms. Improving the rover’s navigation with advanced techniques, like SLAM (Simultaneous Localization and Mapping), could allow it to perform better in tight spaces or on difficult terrain. Additionally, integrating cloud-based storage and more advanced data analytics could provide more effective real-time monitoring and easier management of the collected video footage.

By addressing these limitations and incorporating these improvements, the system has the potential to become even more reliable and adaptable, expanding its usefulness in various real-world security applications.

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APPENDICES

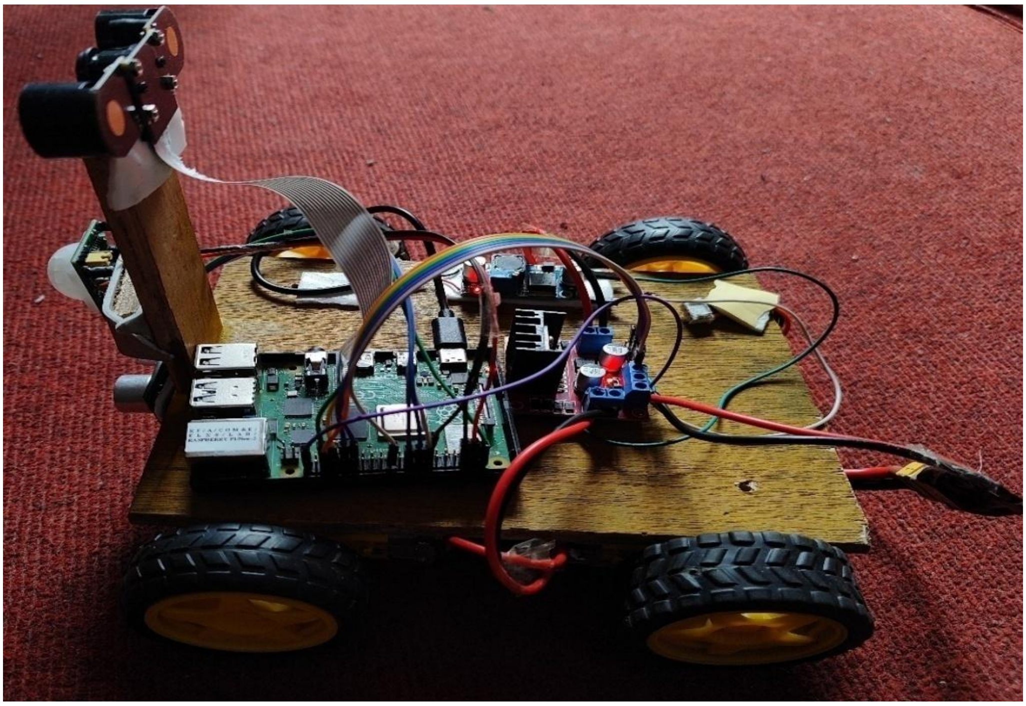

Figure: Side view of Rover

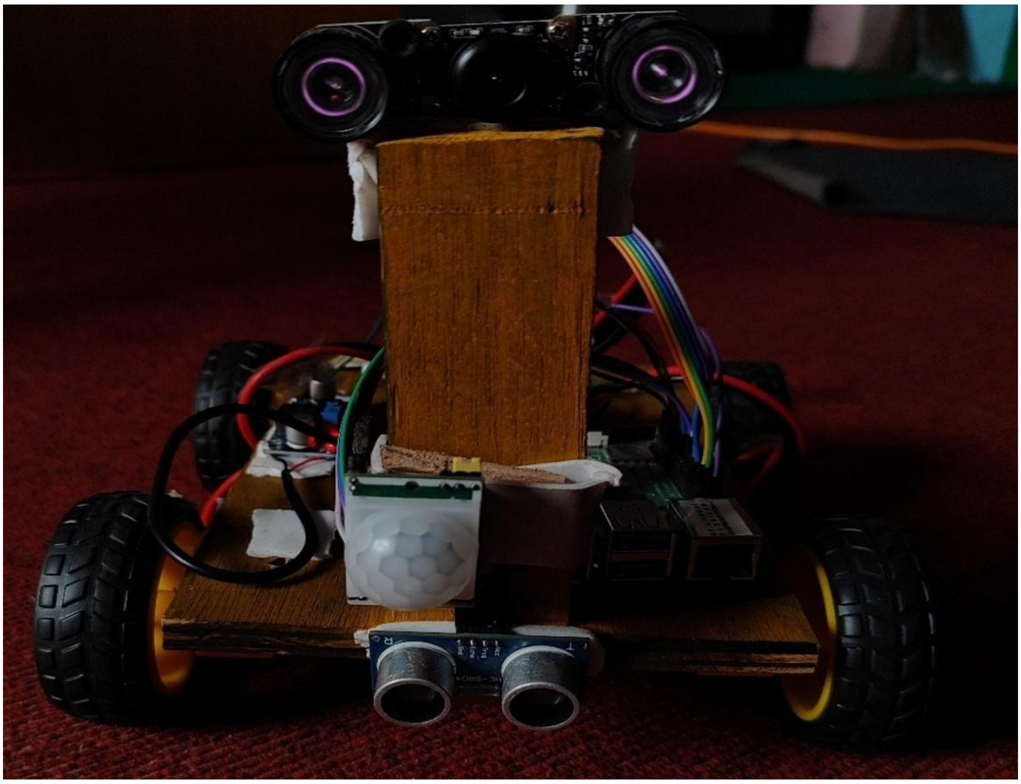

Figure: Front View of Rover

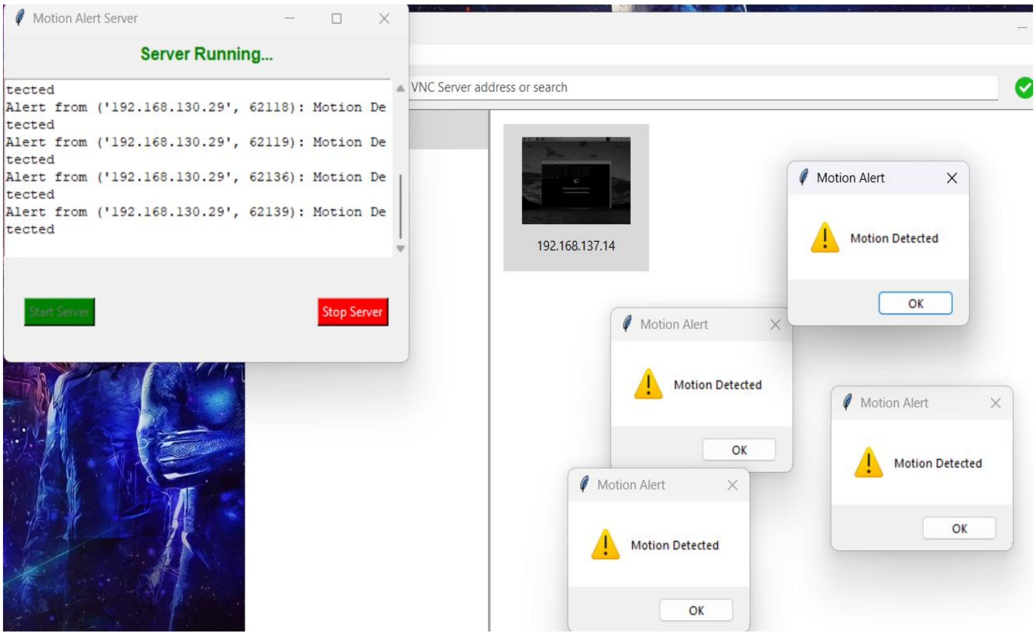

Figure: Motion Alert System

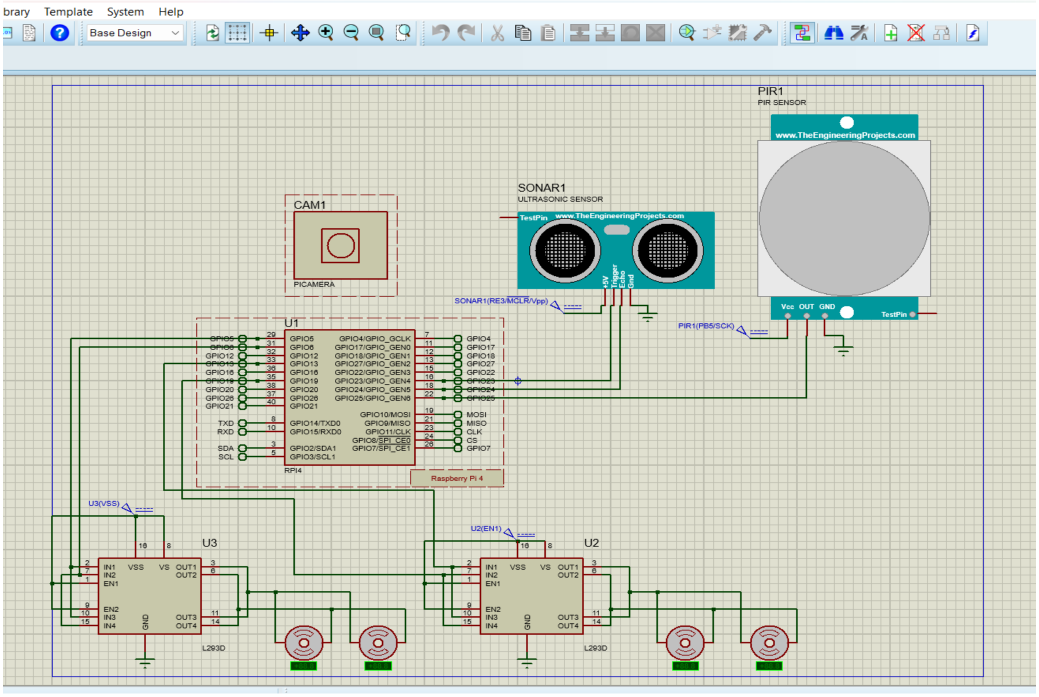

Figure: Proteus Simulation